

Fetal Health Multiclassification Using Cardiotocography Data

Machine Learning Mini-Project | UE24CS352A

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1. Problem Statement

The project studies multiclass classification of fetal health states from cardiotocography (CTG) measurements. The objective is to classify each CTG record as **Normal**, **Suspect**, or **Pathological** using supervised machine learning. The work follows the assigned Stanford CS229 project topic and compares multiple machine learning approaches on the CTG dataset.

2. Dataset

The Cardiotocography dataset was obtained from the UCI Machine Learning Repository. It contains **2,126 records** and **21 predictive features**. The target variable NSP uses 1 = Normal, 2 = Suspect, and 3 = Pathological. UCI reports that the CTGs were classified by three expert obstetricians with a consensus fetal-state label and that the dataset has no missing values.

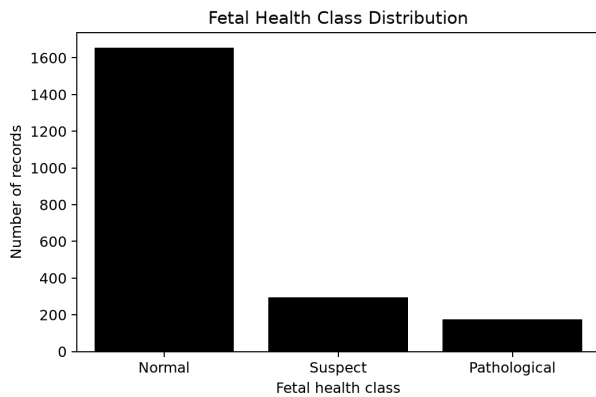


Figure 1. Fetal health class distribution.

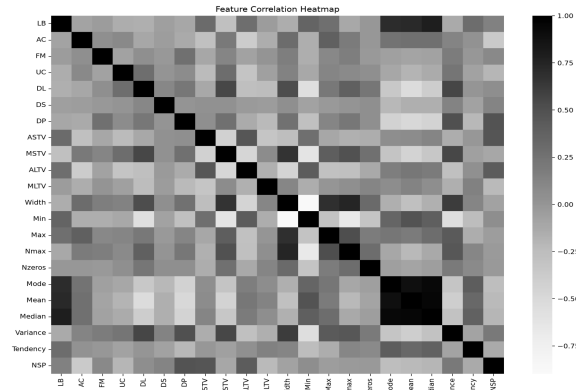


Figure 2. Feature correlation heatmap.

3. Approach and Implementation

A stratified 70:30 train/test split was used with random state 42. Feature scaling was applied to Logistic Regression, Linear SVM, KNN, SVM, and MLP; tree-based models used the original feature values. Ten models were evaluated: Logistic Regression, Linear SVM, KNN, Decision Tree, Random Forest, Gradient Boosting, SVM, MLP, XGBoost, and LightGBM. Accuracy, macro precision, macro recall, macro F1, and weighted F1 were recorded, along with confusion matrices.

4. Exploratory Analysis

The class distribution is strongly imbalanced toward Normal records. The correlation heatmap also shows noticeable relationships among several histogram and variability features. Because of the class imbalance, macro-averaged metrics are useful for judging performance across all three classes.

5. Model Evaluation and Results

LightGBM was the best-performing model, achieving **94.20% accuracy** and **89.49% macro F1**. XGBoost was a close second at 93.89% accuracy and 89.38% macro F1. The strongest three models were LightGBM, XGBoost, and Gradient Boosting.

Model	Accuracy	Macro F1	Weighted F1
LightGBM	94.20%	89.49%	94.00%
XGBoost	93.89%	89.38%	93.68%
Gradient Boosting	93.42%	88.04%	93.19%
Random Forest	92.95%	87.15%	92.76%
Multi-layer Perceptron	92.79%	85.89%	92.60%
Decision Tree	90.75%	84.20%	90.60%
Logistic Regression	89.34%	80.52%	89.47%
Support Vector Machine	90.28%	80.14%	89.91%
Linear SVM	88.56%	78.76%	88.58%
K-Nearest Neighbors	87.62%	75.05%	86.99%

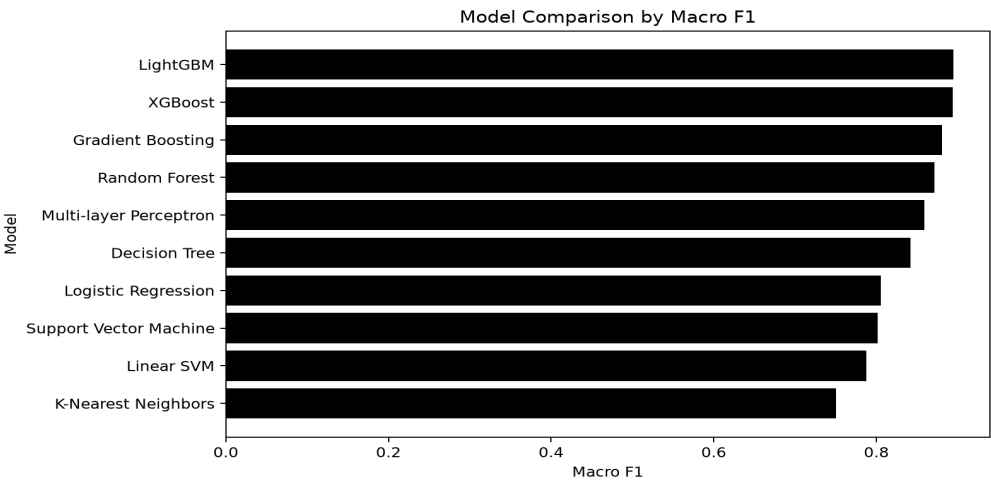


Figure 3. Model comparison using macro F1.

LightGBM - Confusion Matrix			
True label	Normal	Suspect	Pathological
	487	7	3
	21	64	3
	2	1	50
	Normal	Suspect	Pathological
	Predicted label		

6. Error Analysis

LightGBM correctly classified 487 Normal, 64 Suspect, and 50 Pathological test records. The main error pattern was between Normal and Suspect: 21 Suspect records were predicted as Normal. The Pathological class was also separated well, with 50 of 53 records correctly classified. The confusion matrix supports the use of macro F1 alongside accuracy.

Figure 4. LightGBM confusion matrix.

7. Conclusion and Limitations

The experiment successfully compared ten machine learning models for fetal-health multiclassification. LightGBM produced the strongest result, while boosting and ensemble tree models generally outperformed the linear and KNN methods. The project is an academic prototype and **not a clinically validated diagnostic system**. Limitations include class imbalance and reliance on a single stratified train/test split. Future work can include cross-validation, hyperparameter tuning, class-balancing methods, and deeper feature-importance analysis.

References

1. UCI Machine Learning Repository, *Cardiotocography*, Dataset 193, Campos & Bernardes, DOI 10.24432/C51S4N.
2. J. R. Ereyi and M. Huo, *Multiclass Classification of Fetal Health using Cardiotocogram Data*, Stanford CS229 Project, Spring 2021.
3. Project implementation and experimental results produced by the authors.